Coordinated Datacenters as Virtual Power Plants: Demand Flexibility for Transmission-Level Congestion Mitigation

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***Abstract*—The unprecedented growth of large-scale datacenters is intensifying stress on electric power transmission networks and exacerbating congestion challenges for grid operators. This paper investigates the deployment of coordinated datacenter flexibility modeled as a Virtual Power Plant (VPP) to mitigate transmission-level congestion through demand-side control. Using AC power flow simulations on the IEEE 118-bus test system with engineered congestion, five 200 MW datacenters respond to a centralized congestion-based stress signal by temporarily deferring non-critical computational workloads within operational limits. A comprehensive parameter sweep is conducted across datacenter intervention thresholds, stress signal gain, flexibility bounds, placement methods, and signal allocation to identify optimal operating parameters and design principles. Experimental results indicate that intervention timing exhibits an *optimal threshold* at which effectiveness is maximized, congestion stress signals *saturate* at modest levels beyond which additional increases waste deferred energy without improving performance, *spatially-optimized* stress signal allocation schemes significantly outperform uniform signaling, and *strategic datacenter placement* provides modest improvements in efficiency as compared to random placement. Together, these findings demonstrate that, with proper calibration, coordinated datacenters can act as fast, grid-aware assets for transmission congestion mitigation while clarifying the operational constraints that determine demand-side flexibility effectiveness.**

***Index Terms*—Datacenters, demand response, load flexibility, power system congestion, virtual power plant (VPP).**

I. Introduction

AI-driven demand growth from hyper-scale datacenters is rapidly transforming power system planning and operations, with clusters of facilities now rivaling utility-scale generation in size [1]. Recent projections attribute a large share of future electricity demand in the U.S. to AI and cloud workloads, with estimates ranging from 325 TWh to 580 TWh by 2028 which is equivalent to 74–132 GW power demand at 50% capacity utilization. This represents 6.7%–12% of total U.S. electricity consumption and more than doubles the 176 TWh consumed in 2023, as outlined by Shehabi *et al.* [2]. Multi-hundred-megawatt campuses are increasingly being deployed in major transmission hubs, with individual facilities drawing as much power as small cities. At the same time, much of the existing transmission was not designed for such fast, spatially clustered load growth, amplifying congestion risks and stressing legacy infrastructure [3].

These trends are particularly visible in high-growth regions such as Texas, where aggressive datacenter and industrial electrification plans are colliding with long transmission development timelines. Texas datacenter electricity demand is projected to grow by around 88%, i.e., from 7.7 GW in 2025 to 14.5 GW by 2030 [4], with ERCOT load forecasts reaching over 150 GW by 2030 driven by datacenter and industrial electrification [5]. Plans for major transmission expansions illustrate both the recognition of emerging congestion challenges and the reality that large projects often require many years, e.g., 5–20 years for ERCOT's recently approved 765 kV extra-high voltage transmission lines [5], creating a mismatch between the pace of AI/datacenter acceleration and grid innovation. This growing temporal and spatial mismatch raises concerns about increased curtailment, congestion costs, and reliability risks for grid operators, as highlighted by recent documents issued by governing agencies such as the Federal Energy Regulatory Commission [6].

In parallel, datacenters themselves are evolving from passive loads into flexible grid-interactive assets, equipped with on-site backup generation, energy storage, and sophisticated control systems capable of responding rapidly to evolving grid conditions. Major technology companies such as Google have begun implementing demand flexibility programs that defer machine learning workloads to off-peak periods or curtail non-essential operations during grid stress events [7]. These characteristics make datacenters ideal candidates for Virtual Power Plant (VPP) operation, with studies demonstrating 10% load curtailment within a span of 5 minutes through demand-side management [8]. This emerging flexibility opens the door to treating coordinated datacenters as VPPs, where aggregated load adjustments can support system needs such as congestion relief, reserve provision, and voltage support [8]–[10]. However, the operational regimes, control designs, and parameter ranges under which such flexibility materially mitigates transmission congestion remain poorly quantified,

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in particular at the transmission level where power flow interactions are highly network-dependent.

Some existing works have explored datacenter flexibility through workload migration and spatial load balancing, as demonstrated by Liu *et al.* [9], and VPP operation frameworks as reported in [8], [10]. However, these studies focused primarily on the power distribution-level demand response or energy cost optimization, leaving the transmission-level (the backbone of power system) congestion mitigation through coordinated datacenter VPPs unexplored [11].

Consequently, there is a pressing need to systematically characterize how datacenter load flexibility can be harnessed as a fast, grid-aware tool for transmission congestion management. Highly motivated by the challenge associated with congestion mitigation and rise in datacenter load demand, this paper addresses this gap by modeling coordinated datacenter flexibility as a VPP embedded in an IEEE 118-bus test system with AC power flow. The key contributions of this paper lie in the following:

- Demonstration of coordinated datacenter VPP control as an effective grid-stress mitigation technique, achieving near-complete overload elimination (99.4% severity reduction) under baseline congestion;
- Identification of a critical congestion threshold where intervention is optimal (90%), showing that premature and delayed intervention lead to an increase in curtailed energy and decrease in efficiency, respectively;
- Establishment of a congestion stress signal saturation behavior, proving that incentive signals above a gain value ($\kappa \approx 200$) provide no additional congestion relief while consuming more curtailment energy; and
- Spatially-optimized signaling outperforms uniform allocation by reducing curtailment energy by 40%, demonstrating that sensitivity-aware signal distribution significantly improves operational efficiency.
- This paper thus contributes to the area of demand side management by leveraging the evolving datacenters as grid-aware VPP assets for reliable operations of power systems.

The rest of the paper is organized as follows. Section II describes the system model including the IEEE 118-bus transmission network, congestion engineering, datacenter placement, and grid-datacenter coordination framework. Section III presents the simulation setup and evaluation methodology. Section IV presents the experimental results and subsequent analysis, and Section V provides a conclusion and discusses future work directions.

II. Proposed Approach for Coordinated Datacenter Demand Management

A. Transmission Network Model

We employ the IEEE 118-bus transmission test system as the physical layer, consisting of 118 buses, 173 transmission lines, 53 generators, and 99 loads. The network is modeled using AC power flow with full consideration of voltage magnitudes, phase angles, and reactive power flows. All simulations are implemented using the pandapower Python library [12], with convergence verified using the Newton-Raphson method.

Let $\mathcal{L}$ denote the set of all transmission lines in the network, with cardinality $|\mathcal{L}| = L$. To create realistic diurnal congestion patterns, we employ a calibration procedure ensuring overloads occur only during peak demand while lines remain clear during off-peak hours.

We first identify candidate lines exhibiting significant diurnal variation by computing the peak-to-offpeak current ratio:

$$r_\ell = \frac{I_\ell^{\text{peak}}}{I_\ell^{\text{offpeak}}}, \tag{1}$$

where I_ℓ^{peak} and I_ℓ^{offpeak} are currents when system loads are scaled by $m_{\text{peak}} = 1.10$ and $m_{\text{offpeak}} = 0.70$, respectively. Lines with $r_\ell > 1.5$ form the candidate set, from which the $K = 5$ lines with highest peak current are selected.

For each selected line $\ell \in \mathcal{L}_{\text{cong}}$, the thermal limit is calibrated to achieve target loading $\rho_{\text{target}} = 1.20$ at peak:

$$I_\ell^{\text{max,new}} = \frac{I_\ell^{\text{peak}}}{\rho_{\text{target}}}. \tag{2}$$

Since selected lines satisfy $r_\ell > 1.5$, their off-peak loading is guaranteed at or below 80%, ensuring recovery windows exist.

B. Datacenter Placement and Parameterization

We model $N = 5$ geographically distributed datacenters as controllable loads with nominal power consumption $p^{\text{nom}} = 200$ MW per facility. Two placement methods are evaluated:

► **Strategic:** Datacenters are placed at endpoints of the congested transmission lines by ensuring direct electrical proximity to the engineered congestion, thereby maximizing impact of load adjustments.

► **Random:** Datacenters are randomly distributed across existing load buses to represent uncoordinated placement decisions.

Each datacenter i is parameterized by a flexibility coefficient $\phi_i \in [0, 1]$, representing the fraction of nominal demand that can be adjusted. Let P_i^{nom} denote the nominal power consumption of datacenter i. The control action $\Delta P_i(k)$ defines the deviation from nominal demand and is constrained by

$$\Delta P_i(k) \in [-\phi_i P_i^{\text{nom}},\ \phi_i P_i^{\text{nom}}]\,. \tag{3}$$

Negative deviations correspond to load curtailments via workload deferral, while positive deviations represent increased consumption for deferred work.

Each datacenter maintains a deferred-energy state $E_{\text{def},i}(k)$ (MWh) representing the cumulative workload that has been curtailed but not yet recovered. For a timestep

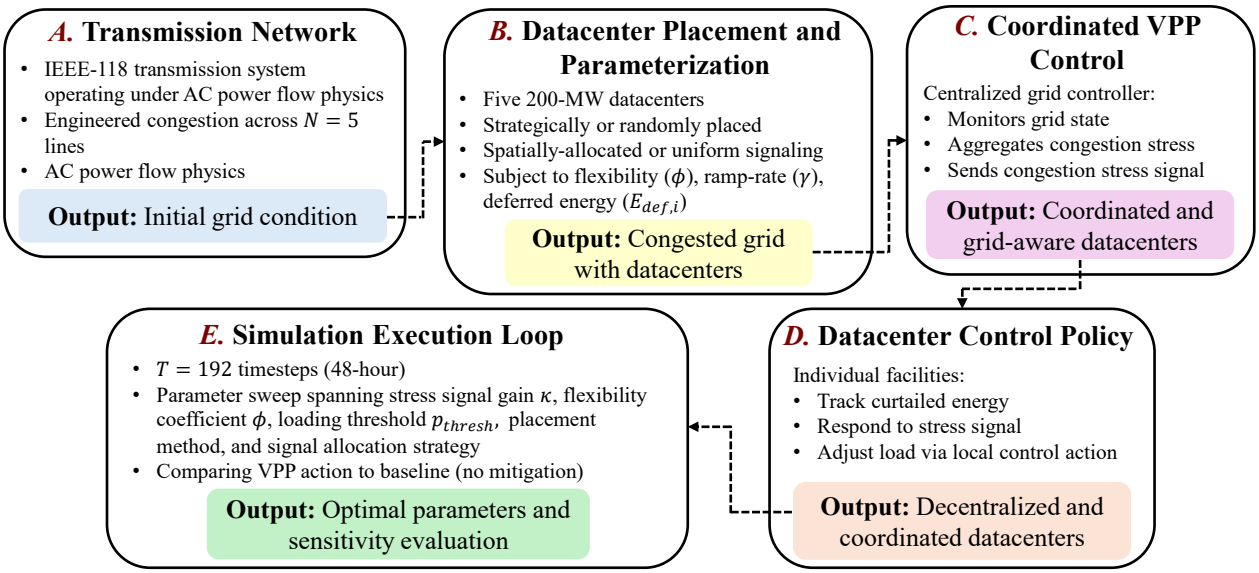

Fig. 1. Proposed grid-aware VPP-based congestion mitigation framework employing flexible-load datacenters. Five 200 MW datacenters receive a centralized congestion stress signal and respond with flexible load actions, which are subject to local constraints and parameters. Performance is evaluated against baseline scenarios through a comprehensive parameter sweep.

duration $\Delta t = 0.25$ corresponding to 15 minutes, the backlog evolves as

$$E_{\mathrm{def},i}(k+1) = \mathrm{clip}(E_{\mathrm{def},i}(k) - \Delta P_i(k)\Delta t,\ 0,\ E_{\mathrm{cap},i}), \tag{4}$$

where $\mathrm{clip}(x,a,b) = \min(\max(x,a),b)$ and $E_{\mathrm{cap},i} = P_i^{\mathrm{nom}} \cdot H_{\mathrm{cap}}$ is the maximum deferral capacity (MWh) with $H_{\mathrm{cap}} = 4$ denoting allowable backlog hours.

Datacenter load adjustments are further subject to asymmetric ramp-rate constraints that reflect operational differences between rapid workload curtailment and slower backlog recovery. For each timestep, a datacenter's control action must follow

$$-\gamma_{\downarrow} P_i^{\mathrm{nom}} \ \le\ \Delta P_i(k) - \Delta P_i(k-1) \ \le\ \gamma_{\uparrow} P_i^{\mathrm{nom}}, \tag{5}$$

where $\gamma_{\uparrow} = 0.20$ limits upward adjustments (recovery) to 20% of nominal demand per timestep, and $\gamma_{\downarrow} = 1.0$ permits rapid curtailment. For the operating range considered, the ramp-down rate is non-binding, and curtailment is limited by the flexibility bound rather than the ramp rate.

C. Coordinated VPP Control Architecture

The coordinated VPP control operates in discrete 15-minute timestep over a 48-hour period, indexed by $k \in \{1, 2, \ldots, T\}$ with $T = 192$. At each time step k, the centralized grid controller performs the following actions:

1) **Monitors grid state**: Solves an AC power flow and computes the thermal loading $L_\ell(k)$ (in percent) for all transmission lines ℓ. The corresponding normalized line loading ratio is defined as

$$\rho_\ell(k) = \frac{L_\ell(k)}{100}, \tag{6}$$

which provides a dimensionless measure of proximity to thermal limits and serves as the primary congestion state variable in the proposed system architecture.

2) **Aggregates congestion stress**: Constructs a scalar congestion measure. Let $\boldsymbol{\rho}(k) = [\rho_1(k), \rho_2(k), \ldots, \rho_L(k)]^\top$ denote the vector of line loading ratios at timestep k. The system-wide congestion stress is defined as

$$\mathcal{S}(k) = \sum_{\ell \in \mathcal{L}} [\max(0, \rho_\ell(k) - \rho_{\mathrm{thresh}})]^p, \tag{7}$$

where ρ_{thresh} is a congestion threshold and $p = 2$ applies a quadratic penalty on critical overloads.

3) **Generates congestion stress signal**: Maps the aggregate stress into a dimensionless congestion penalty signal

$$\pi(k) = \kappa \cdot \mathcal{S}(k), \tag{8}$$

where κ is a tunable gain parameter that controls the strength of the grid-side incentives.

4) **Broadcasts incentives**: Distributes the congestion signal to datacenters. In uniform mode, all facilities receive the same signal $\pi_i(k) = \pi(k)$. In sensitivity-weighted mode, per-datacenter signals are computed as $\pi_i(k) = \pi(k)\, w_i$, where w_i reflects the relative impact of datacenter i on the most congested transmission elements. Sensitivity weights $w_i(k)$ are computed using a finite-difference approximation around the current operating point and normalized such that $\sum_{i=1}^{N} w_i(k) = 1$.

The grid controller does not directly command the actions of the facilities; it only broadcasts signal-based incentives. Individual datacenters independently compute feasible responses subject to local parameters, i.e., flexibility and ramp-rate constraints.

D. Datacenter Control Policy

Each datacenter receives the congestion signal $\pi_i(k)$ and computes a feasible control action $\Delta P_i(k)$ representing the adjustment to its nominal power consumption P_i^{nom}.

The recovery bias $\beta_i(k)$ encodes the incentive to restore deferred workloads and is gated to prevent rebound during congestion. It is defined as

$$\beta_i(k) = \begin{cases} 0, & \max_{\ell \in \mathcal{L}} \rho_\ell(k) > \rho_{\text{thresh}}, \\ \eta \dfrac{E_{\text{def},i}(k)}{\Delta t}, & \text{otherwise}, \end{cases} \tag{9}$$

where Δt represents the timestep duration, and η is a recovery-gain parameter. Across all experiments, $\eta = 1.0$ and is omitted for simplicity.

The control decision is governed by flexibility, backlog, and ramp-rate constraints as described in (3), (4), and (5), respectively. Each datacenter selects its power adjustment by minimizing a local quadratic cost

$$\Delta P_i(k) = \arg\min_{\Delta P} \left(\lambda \, \Delta P^2 + \pi_i(k) \, \Delta P - \beta_i(k) \, \Delta P \right), \tag{10}$$

where $\lambda = 0.05$ is a common effort-penalty coefficient. This yields the closed-form solution projected onto the previously defined feasibility bounds, and it is given by

$$\Delta P_i^\star(k) = \frac{\beta_i(k) - \pi_i(k)}{2\lambda}, \tag{11}$$

E. Simulation Execution Loop

The complete VPP simulation integrates grid state monitoring, control computation, and power flow updates in a closed-loop framework over the 48-hour horizon (192 timesteps with $\Delta t = 0.25$). Prior to grid state evaluation, non-datacenter loads are scaled by a deterministic diurnal profile that creates realistic congestion patterns aligned with typical load demand curves. For each non-datacenter load bus j, the real power demand at timestep k is given by

$$P_j(k) = P_j^{\text{base}} \cdot m\big(h(k)\big), \tag{12}$$

where P_j^{base} is the nominal demand, $h(k) \in [0, 24)$ is the hour corresponding to timestep k, and $m(h)$ is a piecewise load multiplier

$$m(h) = \bar{m} + A \cos\left(\frac{\pi(h - h_0)}{12} \right) \tag{13}$$

where $\bar{m} = 0.90$ is the mean load level, $A = 0.20$ is the diurnal amplitude, and $h_0 = 18$ is the hour of peak demand. This yields a minimum multiplier of 0.70 at $h = 6$ (early morning) and maximum of 1.10 at $h = 18$ (evening peak). At each timestep, the simulation advances through three phases as described below:

Pre-control assessment: The grid state is evaluated under current loading conditions. An AC power flow is solved to obtain line loading ratios $\rho_\ell(k)$ across all lines, establishing a baseline congestion level before VPP intervention.

Coordinated control: The central controller computes the congestion stress signal $\mathcal{S}(k)$ and broadcasts incentives $\pi_i(k)$ to all datacenters. Each facility independently computes its optimal load adjustment ΔP_i by solving a constrained local optimization that balances congestion relief incentives, recovery bias, and effort costs, while respecting flexibility bounds, backlog constraints, and ramp-rate limits. Most importantly, this decentralized decision-making structure preserves datacenter autonomy while achieving system-level coordination.

Post-control evaluation: Datacenter load adjustments are applied to the network model, and the AC power flow is resolved to obtain a post-control grid state. Line loading ratios are recomputed to assess the impact of VPP intervention on transmission congestion. All state variables are logged for subsequent analysis.

This closed-loop structure captures key feedback dynamics between grid congestion, signal-based coordination, and distributed datacenter response, enabling systematic evaluation of VPP effectiveness across diverse operating conditions.

III. Simulation Setup and Evaluation Metrics

A. Parameter Sweep Methodology

To ensure fair comparisons across control configurations and isolate the impact of individual parameters on VPP effectiveness, we employ a *frozen-template* sweep methodology. This approach decouples grid topology from control parameters by fixing the physical network configuration for groups of scenarios that differ only in their control settings. For each unique combination of grid configuration parameters, we construct a scenario template by:

1) Building the base IEEE 118 network,
2) Applying congestion engineering to identify and tighten the top-K most loaded lines, and
3) Selecting datacenter bus locations according to the specified placement strategy.

All parameter combinations sharing the same template are evaluated on identical grid configurations, with variations limited to control parameters such as stress signal gain κ, intervention threshold ρ_{thresh}, and flexibility coefficient ϕ. This eliminates variation in line congestion and datacenter siting that would otherwise risk integrity in results. For each VPP scenario with $\phi > 0$ (which allows for VPP intervention), a corresponding baseline simulation with $\phi = 0$ (no control) is executed on the same template, enabling direct computation of congestion reduction metrics.

TABLE I
Parameter Sweep Configuration

Parameter	Range/Values
Flexibility coefficient, ϕ	0.15, 0.25, 0.35, 0.45
Signal gain, κ	200, 400, 600, 800
Intervention threshold, ρ_{thresh}	0.90, 0.92, …, 1.00
Signal mode	uniform, weighted
Placement	strategic, random
Fixed: $T = 192$, $\lambda = 0.05$, $P_i^{nom} = 200$ MW, $N = 5$, $H_{\text{cap}} = 4$, $\gamma_\uparrow = 0.20$, $\gamma_\downarrow = 1.0$	

B. Parameter Space

The experimental design consists of three complementary sweeps that systematically isolate key control parameters:

Initial characterization sweep: explores joint parameter space to identify promising operating regimes across flexibility coefficients, stress signal scaling gain, intervention thresholds, signaling strategies, and datacenter placement methods (strategic, random).

Threshold validation sweep: focuses on intervention timing by testing $\rho_{\text{thresh}} \in \{0.90, 0.92, \ldots, 1.00\}$ with optimal settings ($\phi = 0.45$, $\kappa = 200$, strategic placement, weighted signaling).

Signal saturation sweep: characterizes the response to incentive magnitude by testing $\kappa \in \{100, 200, \ldots, 1000\}$ with optimal threshold (ρ_{thresh}) and other parameters fixed. Table I summarizes the parameter ranges explored. All sweeps employ the frozen-template methodology to ensure fair comparisons.

C. Performance Metrics

System performance is quantified over the 192-timestep horizon using the following metrics derived from line loading ratios $\rho_\ell(k)$ and control actions $\Delta P_i(k)$. The metrics utilized in this paper to evaluate parametric framework performance are described below.

Overload duration: The number of timesteps for which at least one line exceeds its thermal limit threshold,

$$N_{\text{overload}} = \sum_{k=1}^{T} \mathbb{1}\left\{ \max_{\ell \in \mathcal{L}} \rho_\ell(k) > 1.0 \right\}, \tag{14}$$

where $\mathbb{1}\{\cdot\}$ denotes the indicator function.

Overload severity: The cumulative excess loading beyond thermal limits,

$$\mathcal{M}_{\text{sev}} = \sum_{k=1}^{T} \sum_{\ell \in \mathcal{L}} \max(0, \rho_\ell(k) - 1.0)\,. \tag{15}$$

Energy curtailed: The total net energy curtailed by the facilities,

$$E_{\text{curtail}} = \sum_{k=1}^{T} \max\left(0,\ -\sum_{i=1}^{N} \Delta P_i(k)\right) \Delta t. \tag{16}$$

Recovery ratio: The fraction of curtailed energy that is subsequently recovered,

$$R_{\text{rec}} = \frac{E_{\text{recover}}}{E_{\text{curtail}}}. \tag{17}$$

Values approaching 1.0 indicate effective workload deferral with minimal permanent loss, while low values suggest curtailed work cannot be recovered within the simulation horizon.

Severity reduction: The percentage reduction in cumulative overload severity achieved by VPP control relative to baseline,

$$\Delta_{\text{sev}} = \frac{\mathcal{M}_{\text{sev}}^{\text{base}} - \mathcal{M}_{\text{sev}}^{\text{VPP}}}{\mathcal{M}_{\text{sev}}^{\text{base}}} \times 100\%, \tag{18}$$

where $\mathcal{M}_{\text{sev}}^{VPP}$ represents the severity under VPP control.

TABLE II
OPTIMAL VPP CONFIGURATION AND BASELINE COMPARISON

Metric	Baseline	VPP
Overload steps	75	10
Overload severity	37.3	0.2
Severity reduction	—	99.4%
Curtailment (MWh)	0	4,693
Recovery (MWh)	—	2,430
Recovery ratio	—	0.52

Optimal: $\phi = 0.45$; $\kappa = 200$; $\rho_{thresh} = 0.90$, strategic, weighted.

TABLE III
IMPACT OF PARAMETER VARIATIONS FROM OPTIMAL CONFIGURATION

Parameter	Value	Reduction (%)	Recovery	Curtailed (MWh)	ΔCurtailed
Optimal configuration		99.4	0.52	4,693	—
Flexibility, ϕ	0.20	66.1	0.52	2,471	−47%
	0.30	88.0	0.52	3,442	−27%
Signal gain, κ	100	98.8	0.52	3,539	−25%
	500	99.4	0.51	5,592	+19%
	800	99.2	0.51	5,849	+25%
Threshold, ρ_{thresh}	0.85	99.9	0.51	5,495	+17%
	0.95	96.5	0.55	3,754	−20%
	1.00	83.4	0.63	2,685	−43%
Placement	random	99.3	0.52	5,191	+11%
Signal	uniform	99.4	0.51	7,822	+67%

ΔCurtailed = percent change in curtailment relative to optimal.
Optimal: $\phi = 0.45$; $\kappa = 200$; $\rho_{thresh} = 0.90$, strategic, weighted.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the simulation results evaluating the performance of the proposed coordinated datacenter VPP framework for mitigating transmission-level congestion. A comprehensive frozen-template parameter sweep across intervention thresholds, signal gains, flexibility coefficients, placement strategies, and signal allocation methods was conducted using AC power flow simulations on the IEEE 118-bus test system. Results are organized first by presenting the optimal parameters for the proposed framework, followed by individual parameter sensitivity analysis and evaluation of their effects on congestion mitigation performance.

Table II presents the performance of the optimal VPP configuration compared to the uncontrolled baseline. Under baseline conditions (no VPP intervention), the system experiences 75 overloaded timesteps with cumulative severity of 37.3. With coordinated VPP control (using optimal parameters), overloaded timesteps are reduced to 10 and severity drops to 0.2, representing a 99.4% reduction in overload severity. The VPP achieves this by curtailing 4,693 MWh of datacenter load during congested periods, of which 2,430 MWh (52%) is recovered during subsequent off-peak hours.

Table III quantifies the impact of deviating from the optimal configuration by varying one parameter at a time. Several key insights emerge as described below.

Flexibility coefficient (ϕ): Reducing flexibility from 0.45 to 0.20 causes severity reduction to drop from 99.4% to 66.1%, indicating that insufficient flexibility renders the VPP ineffective. A minimum flexibility of $\phi \geq 0.30$ is required to achieve reduction above 85%.

Signal gain (κ): The congestion signal exhibits satura-

tion behavior. Increasing κ from 200 to 800 provides no improvement in severity reduction (99.4% to 99.2%) while increasing curtailment by 25%. Conversely, $\kappa = 100$ achieves 98.8% reduction with 25% less curtailment, suggesting modest congestion stress signals are sufficient.

Intervention threshold (ρ_{thresh}): Earlier intervention ($\rho_{\text{thresh}} = 0.85$) achieves marginally better reduction (99.9%) but increases curtailment by 17%. Later intervention ($\rho_{\text{thresh}} = 1.00$) reduces curtailment by 43% but only achieves 83.4% severity reduction, as VPPs respond too late to prevent overloads. The optimal threshold $\rho_{\text{thresh}} = 0.90$ balances effectiveness and efficiency, meaning it is optimally proactive to line violations.

Placement strategy: Strategic placement near congested lines achieves comparable reduction to random placement (99.4% vs 99.3%) but with 10% less curtailment, demonstrating efficiency gains from coordinated siting.

Signaling mode: Weighted (sensitivity-based) signaling achieves identical reduction to uniform allocation (99.4%) but with 40% less curtailment (4,693 MWh vs. 7,822 MWh). This result indicates that spatially-aware congestion signal allocation significantly improves efficiency by directing curtailment to datacenters with highest grid impact.

The obtained results clearly show the precision required in designing coordinated grid-aware congestion mitigation frameworks. Most importantly, the optimal parameters allow the proposed VPP control framework to significantly mitigate transmission-level congestion while allowing for recovery during off-peak periods.

V. Conclusion

This paper presented the use of coordinated datacenter demand flexibility, operating as a VPP, to mitigate transmission-level congestion through systematic AC power flow simulations on the IEEE 118-bus system. Through comprehensive parameter sweeps across parameters such as the flexibility coefficient ϕ, congestion signal gain κ, intervention threshold ρ_{thresh}, placement methods and signal strategies, this paper identified critical operational domains that determined VPP effectiveness for transmission-level congestion mitigation.

Three critical system design parameters emerged from the analysis. *First,* intervention timing (controlled by parameter ρ_{thresh}) exhibited an optimal value of 0.9, where premature intervention ($\rho_{thresh} < 0.85$) increased curtailment energy without proportional benefit, while delayed response ($\rho_{thresh} > 0.95$) reduced curtailed energy and allowed for quick backlog recovery, but exhibited a decrease in severity reduction. *Second,* congestion stress signals saturated at modest gain levels ($\kappa \approx 200$), beyond which additional signal gain provided no improvement in congestion relief while curtailing substantially more energy. *Third,* sensitivity-weighted signal strategies outperformed uniform allocation by reducing required curtailment by 40%, suggesting that spatially-aware signal distribution significantly improved operational efficiency. Importantly, this experiment revealed that strategic datacenter placement near congested corridors yielded only modest efficiency gains (10% reduction in curtailment) compared to random placement, indicating that network-wide coordination could compensate for suboptimal geographic deployment of datacenters. This finding suggested that grid-responsive control could partially compensate for poorly-planned or suboptimal siting decisions.

Overall, by deploying five 200 MW coordinated flexible-load datacenters, the optimal configuration presented in this paper was capable of reducing overload severity by 99.4% through curtailment of 4,693 MWh and the recovery of 2430 MWh over the 48-hour test period. Additionally, VPP-control contained transmission-wide congestion to 10 timesteps, while the baseline conditions experienced 75 congestion timesteps. Parameters ρ_{thresh} (intervention threshold), ϕ (flexibility coefficient), κ (signal gain), datacenter placement strategy, and signal allocation all showcased an impact in the proposed framework's performance.

These findings highlighted coordinated datacenter demand flexibility as a viable transmission-level congestion management tool when properly calibrated, while clarifying the operational constraints, design principles, and parameters that determined effectiveness. Future work could be interesting to extend this framework to include stochastic load and generation variability, multi-area coordination, integration with energy storage and on-site assets, and validation using real datacenter work traces.

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